



KOLEJ MATRIKULASI MELAKA

COMPUTER SCIENCE 1
IQUA
DC014
SEMESTER I, SESSION 2023/2024

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NAME : _____
MATRIC NO. : _____
TUTORIAL : _____

Answer all questions.

- 1** Delima College is looking to enhance its online learning experience by integrating multimedia elements into its course materials. The college aims to provide engaging and interactive content to its remote students, offering a comprehensive and dynamic learning environment.

- (a) Describe **one (1)** multimedia element that can be used for the above situation.

[2 marks]

- (b) Identify the most suitable multimedia element based on the given purpose.

[2 marks]

	Purpose	Multimedia Element
(i)	Captivates and maintains audience attention more effectively than static content, making it a powerful tool for increasing user engagement and retention.	
(ii)	Enables the simulation and visualization of scenarios, processes, and simulations that are difficult or impossible to replicate in real life.	

- (c) Distinguish two-dimensional (2D) and three-dimensional (3D) multimedia application representation below.

[2 marks]

Criteria	Two-dimensional (2D) design	Three-dimensional (3D) design
Dimension		



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- 2 Designing a multimedia project involves a systematic approach to create engaging and effective multimedia content.

- (a) Identify and describe the phase based on the given activities.

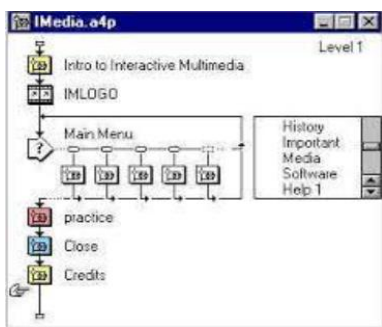
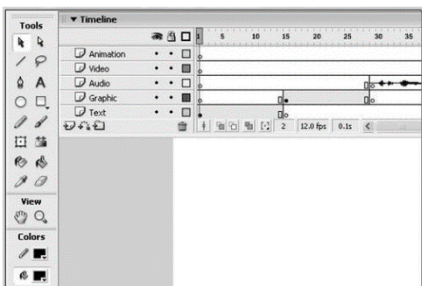
[6 marks]

	Activities	Phase	Description
(i)	Models of artwork are created, rendered, and then prepared for use in the authoring tool.		
(ii)	Writing the final Multimedia products in suitable platform.		

- (b) Multimedia authoring tools are software applications that enable users to create interactive and engaging multimedia content.

Differentiate **two (2)** types of authoring tools based on the given table.

[4 marks]

	Image	How multimedia elements are presented?
(i)		
(ii)		

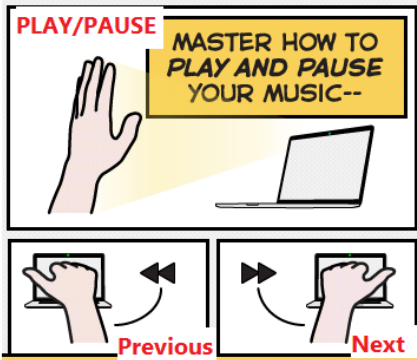


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- (c) Multimedia navigation refers to the process of guiding users through interactive multimedia content, enabling them to access, explore, and interact with various media elements within a digital environment.
Describe multimedia navigation based on the given image.

[2 marks]

Image	Description
	

- (d) Multimedia delivery refers to the process of distributing and presenting multimedia content to users through various digital platforms and channels.

ABC College is organizing a virtual multimedia research symposium to provide students with an opportunity to present their research findings and projects to a wider audience.

- (i) Suggest the most suitable type of multimedia delivery for the above scenario.

[1 mark]

- (ii) Give **one (1)** example for the answer mentioned in 2(d)(i).

[1 mark]

- (iii) Explain **one (1)** disadvantage of medium delivery mentioned in 2(d)(i).

[2 marks]



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- 3 Database systems and file processing systems serve different purposes and have distinct use cases.

- (a) Differentiate file processing system with database system based on the criteria given.

[4 marks]

Criteria	File Processing System	Database System
Data Redundancy		
Data Sharing		

- (b) Describe **one (1)** disadvantage of using database systems.

[2 marks]

- 4 Databases are used in a wide range of applications and industries to efficiently store, manage, and retrieve structured data.

- (a) Rayqal is the database administrator for a medium-sized retail chain with multiple stores across the country. One of your major challenges is ensuring that all stores have real-time access to inventory levels and sales data, while also maintaining data security and integrity.

Explain the concept of data integrity.

[2 marks]

- (b) Identify the most suitable database object based on the description given below.

[2 marks]

	Description	Database Object
(i)	Design an interactive user interface that allows users to view and update customer information in the database.	
(ii)	Generate a summary document that provides a consolidated view of the monthly sales data from the database.	



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- 5 As the backbone of modern communication, networks enable the seamless exchange of information and resources, facilitating collaboration, data sharing, and connectivity on a global scale.

- (a) Elaborate how network is used for Global Positioning System (GPS).

[2 marks]

- (b) List **two (2)** popular GPS application in Malaysia.

[2 marks]

- (c) Identify the type of unbounded networking media based on the given explanation.

[5 marks]

	Explanation	Type of unbounded networking media
(i)	Transmission is used with cell phones and is sent and received through a cellular tower to a tall metal tower with an antenna on top.	
(ii)	This wireless type is not commonly used in modern computer networking. Instead, it has found application in various other areas, such as remote control devices, short-range communication, and data transfer between devices in proximity.	
(iii)	High-frequency radio signals that can send large quantities of data at high speeds over long distances.	
(iv)	This transmission plays a central role in contemporary wireless networks, forming the foundation of mobile communications and enabling widespread connectivity for mobile devices.	
(v)	Communication relays by receiving signals from a transmitting station, amplifying them, and then retransmitting them to a receiving station. This technology is commonly used for long-distance data and television broadcasting.	



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- (d) Identify the type of bounded networking media based on the given characteristic.

[3 marks]

	Characteristic	Type of bounded networking media
(i)	This bounded networking media have a smaller diameter allowing for higher cable density and easier installation in narrow spaces.	
(ii)	The cheapest and easiest cable to connect in a network that connects many computers in one network.	
(iii)	This cable is usually used to transmit television signals; however, it needs a cable modem to access the Internet.	

- (e) Identify the most suitable network device based on the given function below.

[6 marks]

	Function	Network Device
(i)	This device facilitates the conversion of digital signals from a computer or network device into analog signals that can be transmitted over telephone lines.	
(ii)	Provide wireless connectivity to devices within its coverage area, allowing them to connect to a wired network without the need for physical cables.	
(iii)	Accommodate the expansion of the network by providing additional ports for connecting more devices.	
(iv)	Enable a computer or device to connect to a network and communicate with other devices over the network.	
(v)	Responsible for forwarding data packets along the most efficient paths to their intended destinations.	
(vi)	This device will boosting the signal strength and enable the extension of network coverage over longer distances.	



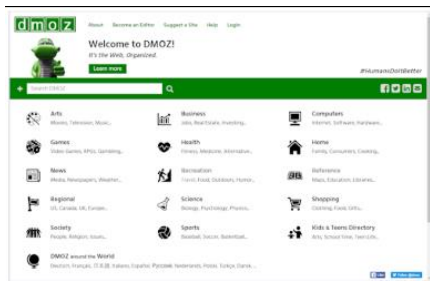
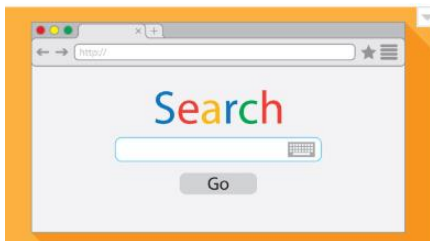
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- 6 Internet searching, also known as web searching, refers to the process of using Internet to locate and retrieve information from the vast resources available on the World Wide Web.

- (a) Identify type of search tools and **one (1)** example based on image given below.

[4 marks]

	Image	Type of Search Tools	Example
(i)			
(ii)			

- (b) Explain **one (1)** of the search tools mentioned in question 6(a).

[2 marks]

- (c) Identify the correct search strategies based on given scenario.

[3 marks]

	Scenario	Search Strategies
(i)	Luhur wants to find books or movies that have the word "star" in the title but not "wars" or "trek" from the Internet.	
(ii)	Vannes prefer to use this style of search strategy by typing more than one keyword in a keyword search box.	
(iii)	Ah Moi type site:melakahariini.my to search the specific news website by using Google search.	



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- (d) Web activities refer to various actions or tasks that users perform online, often through web browsers or web-based applications.

Identify the types of web activities based on the situations given below.

[5 marks]

	Situation	Types of web activities
(i)	Ahmad is an enthusiastic online gamer who uses a gaming platform to play multiplayer video games with friends and strangers from around the world. He enjoys competing in various game genres, teaming up with friends, and participating in online gaming communities to enhance his gaming experience.	
(ii)	Puteri owns a small restaurant and uses an online food delivery service platform to expand her customer reach. Customers can browse her menu, place orders, and have food delivered to their doorstep through the app. Puteri also benefits from features like customer reviews and ratings, helping her improve her business based on feedback from her online customers.	
(iii)	Sophia is a stock market investor, and she regularly checks financial websites to stay updated on the latest developments in the financial markets. She relies on these sources to track the performance of her investments, monitor economic trends, and make informed decisions about buying or selling stocks and other financial assets.	
(iv)	Aqil is a software developer who uses professional networking platforms like LinkedIn to connect with colleagues, peers, and potential employers. He shares his professional achievements, stays informed about industry trends, and joins relevant groups and discussions to expand his professional network and advance his career.	
(v)	Elena is passionate about learning a new language. She uses online language learning platforms, such as Duolingo or Rosetta Stone, to take interactive language courses. She practices reading, writing, speaking, and listening to improve her language skills, and she can track her progress and receive feedback through these online platforms.	



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- (e) Internet filtering can be employed to enhance privacy by managing the type of content that users can access.

- (i) Describe about web browsing privacy.

[2 marks]

- (ii) State **two (2)** tips to improve e-mail privacy.

[2 marks]

- (iii) The Fourth Industrial Revolution (4IR) begins in the early 2010s.
Suggest **one (1)** positive impact of 4IR in the education field.

[2 marks]

END OF QUESTIONS

